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SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

JULY 2023

Course Code: CIV101

Course: **BASIC CIVIL ENGINEERING**

QP No. :UF005-2

Duration: 3 hours

Max. Marks:100

PART-A

Answer all the questions

10 x 2 = 20 Marks

1. What are the four main compounds in Portland cement referred as Bogue's compound? Which Bogue's compound in cement is responsible for early strength development?
2. How does quarrying differ from dressing in the process of obtaining stones for construction?
3. In the context of aggregate, how do fine aggregate and coarse aggregate differ from each other?
4. How would you define a foundation in the context of construction? In what ways does a foundation help to provide stability to a building?
5. You are inspecting a brick wall and notice that there are no closers or bats used in its construction. What might be the implications of this for the stability of the wall?
6. Imagine you are designing a building layout. What are some factors you would consider when deciding where to place the doors?
7. You are a land surveyor tasked with mapping out a new development site. How would you apply the principle of chain surveying to accurately measure the distances and angles between control points on the ground?
8. Let us consider 4 survey lines in all 4 quadrants having whole circle bearing as given below. Convert the given WCB into RB.
Line WCB

AB	65°15'
BC	127°45'
CD	242° 0'
DE	305°30'

9. A dam is 25 m wide and 15 m high. The design flood level for the river is 14 m. What is the required freeboard for this dam to provide adequate safety against overtopping or overflowing during a flood event?
10. What are the key factors that determine the best location for watershed management?

PART -B

Answer all the questions

4 x 15 = 60 Marks

11. You have been tasked with designing a new community centre for a growing town. The centre will need to accommodate a variety of activities, including sports, cultural events, and meetings. The site is located on a flat piece of land with limited space and a tight budget. How do you approach the site selection process to ensure that the community centre is functional, accessible, and attractive? What principles of building planning will guide your decisions?
- (OR)
12. You are a civil engineer working on a project to construct a new skyscraper in a coastal city. The building will be subjected to high winds and salty air, which can cause corrosion and deterioration of concrete over time. You need to select the appropriate type of concrete for the project that will withstand these environmental factors and ensure the safety and durability of the building. What types of concrete would you consider for this project and why? What are the advantages and disadvantages of different types of concrete in this scenario?
 13. You are a geotechnical engineer tasked with designing the foundation system for a new high-rise building in an urban area with poor soil conditions. The building will be constructed on a site with soft clay soil and a high-water table, making it challenging to provide adequate support for the structure. What type of foundation would be

appropriate for this type of soil condition, and explain any five types with neat drawings?

(OR)

14. a) You are an architect working on the design of a new 30-story office building in a busy urban area. One of the key design considerations for the building is the vertical circulation system, which needs to efficiently move people and goods throughout the building while also providing a comfortable and safe experience for users. What design factors would you need to consider, such as the number of stairs required, their size and type, and their location within the building? Illustrate any three with neat sketches. (10)

b) What is the difference between right of way and carriage way in the design and construction of roads and highways? Provide a brief explanation of each concept and explain how they differ. (5)

15. A construction company is levelling a site to prepare it for building a new house. The surveyor has set up the levelling instrument at a benchmark and is taking readings on a staff at different points on the site to determine their R.L. with respect to the benchmark. Use the Rise and Fall method for determining the R.L. of points on a site and do the necessary check. The staff readings are (in meter) 2.0, 2.5, 3.1, 3.9, 0.6, 1.3, 2.0, 2.8, 2.1, 3.1, 0.9, 3.0, 1.2, 2.3, 0.8. The instrument was shifted after 4th, 9th, 11th and 13th reading. The first reading was taken on BM of RL 100 m. Calculate RL of all points. 21

(OR)

16. a) The following offsets were taken at 15 m intervals from a survey line to an irregular boundary line. 3.50, 4.30, 6.75, 5.25, 7.50, 8.80, 7.90, 6.40, 4.40, 3.25 m. Calculate the area enclosed between the survey line, the irregular boundary line by: a) Trapezoidal rule, b) Simpson's rule. (10)

b) How can remote sensing data be processed and analyzed to generate meaningful information about land cover, climate patterns, and other phenomena? And write some of the key applications of remote sensing in fields like agriculture, forestry, and urban planning? (5)

17. a) A farmer in a drought-prone area is considering different irrigation methods to ensure that their crops receive sufficient water during the growing season. What are some of the different methods of irrigation available to the farmer, and how might they weigh the advantages and disadvantages of each approach given the specific conditions of their farm? (8)
- b) Discuss the different types of rainwater harvesting structures and their respective benefits and drawbacks. (7)

(OR)

18. You are a civil engineer working on a hydroelectric power project. Your team is tasked with designing a rigid dam to support the project. Describe the components and types of rigid dams, and explain which type is most suitable for a hydroelectric power project. Additionally, what are some of the potential social and economic benefits and drawbacks of building a dam?

PART-C

Answer the following

1 x 20 = 20 Marks

19. A homeowner is remodeling their kitchen and needs to select materials for the countertops and walls. After completion of the kitchen remodeling, the homeowner is planning to renovate an existing residential building to improve strength and safety, including the replacement of existing wooden beams and columns with concrete elements.
- a) What are the key factors that the homeowner should consider when selecting materials for the kitchen countertops and walls? (10)
- b) What are some of the key considerations that should be kept in mind when selecting and designing new beams, columns, and roof slab for the renovation project? (10)
